

Assignment: Day 2

TCP/IP Suite

Q1. Open Wireshark, capture network traffic while you refresh the Flipkart homepage, and identify one packet each belonging to the Application, Transport, Internet, and Network Access layers by examining their headers.

Ans: I started a capture on my Wi-Fi interface, refreshed the Flipkart homepage, and stopped the capture. Expanding each packet in the Packet Details pane showed its headers stacked layer by layer (Ethernet, IP, TCP, and the application protocol on top). I picked one packet for each TCP/IP layer:

TCP/IP layer	Packet chosen	Header fields examined
Application	DNS Standard query A www.flipkart.com	DNS header: Transaction ID, Flags, Questions. The Queries section shows the name www.flipkart.com being looked up.
Transport	TCP 51322 to 443 [SYN]	TCP header: Source Port 51322, Destination Port 443, Sequence Number 0, SYN flag set, Window Size.
Internet	IPv4 Laptop to Flipkart server	IP header: Source IP (my laptop's private IP), Destination IP (Flipkart server), TTL, Protocol = TCP (6).
Network Access	Ethernet II Frame carrying the IPv4 packet	Ethernet header: Source MAC (my laptop), Destination MAC (Wi-Fi router), Type = IPv4 (0x0800).

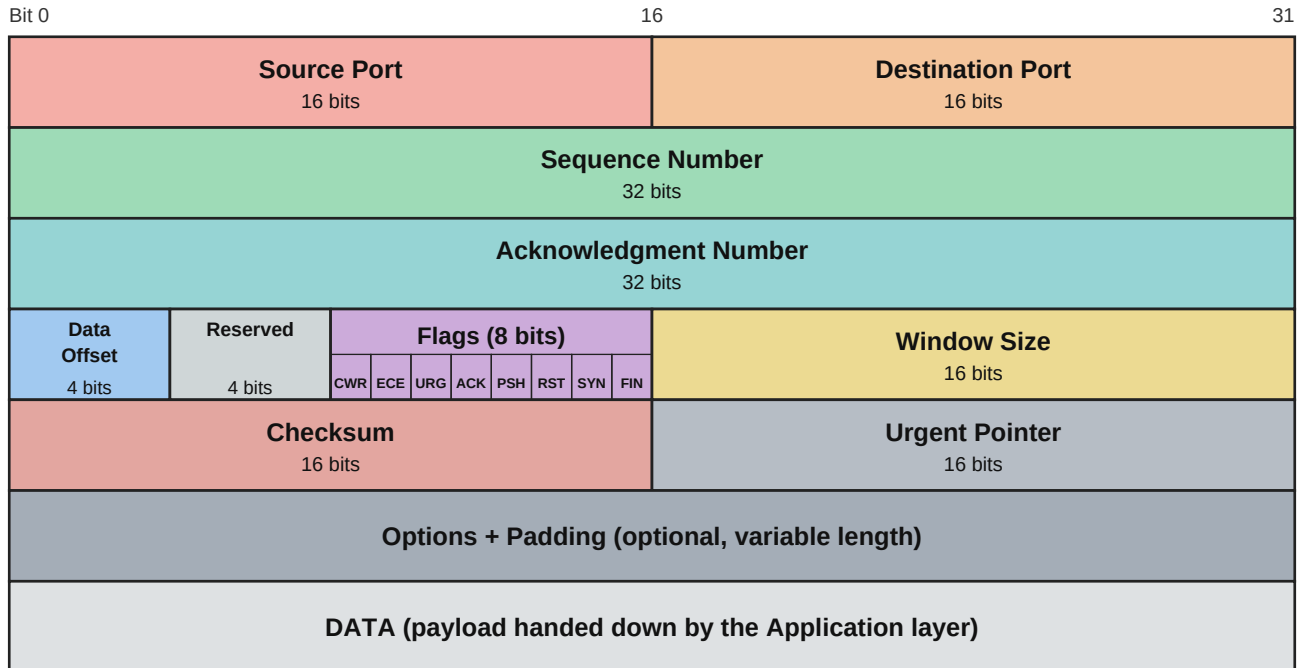
Q2. Create a table comparing the OSI and TCP/IP models, listing all layers side-by-side, and for each layer, give a real-world example from any app you use.

Ans:

OSI layer	Example (OSI)	TCP/IP layer	Example (TCP/IP)
7. Application	Zomato app showing restaurants; WhatsApp chat	Application	Placing a Zomato order or sending a WhatsApp message (HTTPS, DNS)
6. Presentation	Card details encrypted (TLS); Instagram photo in JPEG format		
5. Session	Staying logged in to Zomato while you order		
4. Transport	TCP splits the order into segments and makes sure all arrive	Transport	TCP or UDP carries the order data (port 443)
3. Network	IP address routes the order to Zomato's server	Internet	IP finds the route to the Zomato server
2. Data Link	Phone talks to the Wi-Fi router using its MAC address	Network Access	Wi-Fi or Ethernet frames and the signals that carry them
1. Physical	Wi-Fi radio signals, LAN cable, fibre broadband		

Q3. Given a screenshot of a TCP segment header, label the fields that belong to the Transport layer and explain what each field does in 1-2 lines.

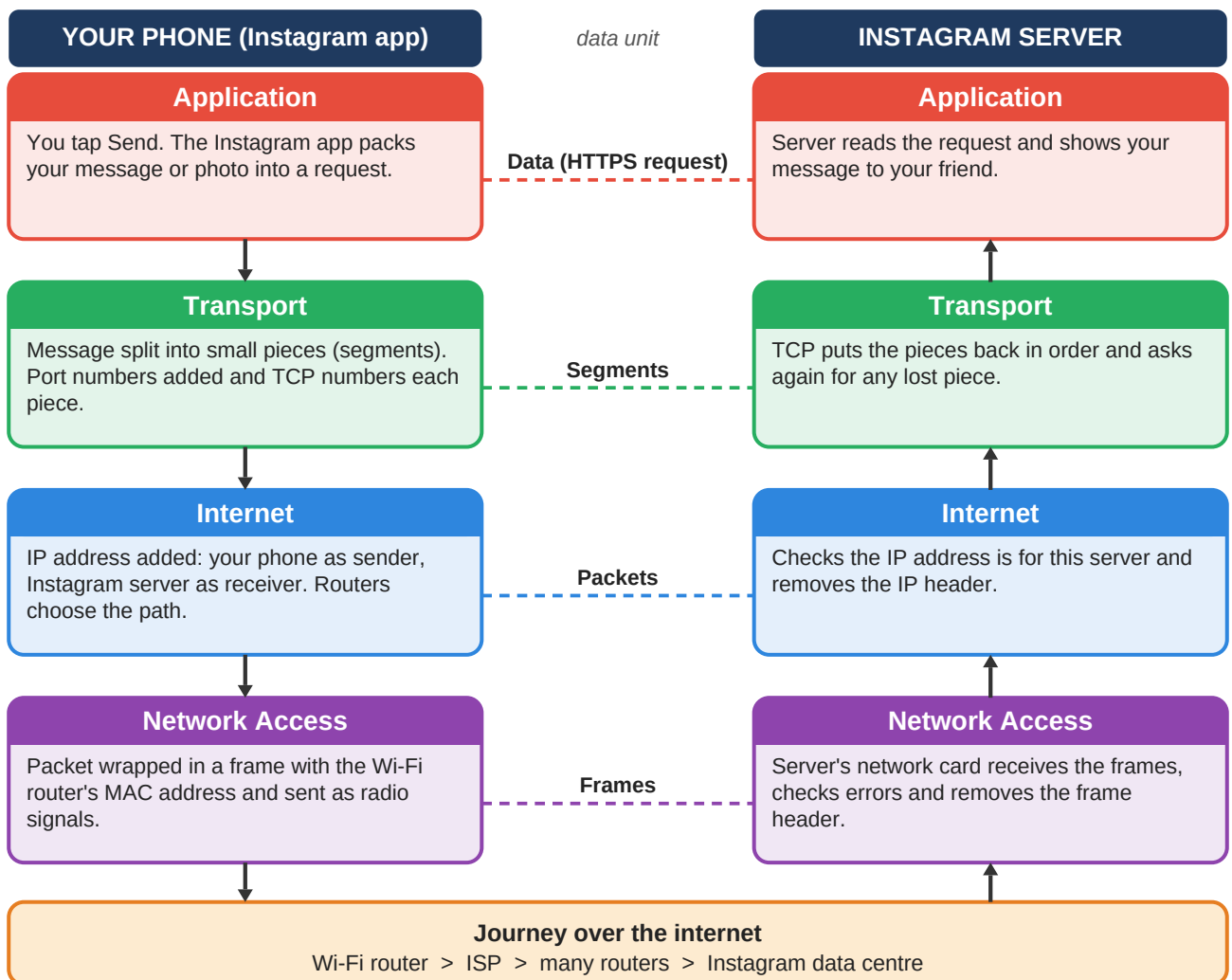
Ans: All the fields of the TCP header belong to the Transport layer. They are labelled below.



Field	Bits	What it does
Source Port	16	Port number of the sending app on my device. It tells the server where to send the reply.
Destination Port	16	Port number of the app on the server, for example 443 for HTTPS or 80 for HTTP.
Sequence Number	32	Number of the first byte in this segment. The receiver uses it to arrange segments in the correct order.
Acknowledgment Number	32	The next byte the receiver expects. It confirms that the earlier data arrived safely.
Data Offset	4	Length of the TCP header. It shows where the actual data starts.
Reserved	4	Kept for future use and normally set to 0.
Flags	8	Control bits: SYN starts a connection, ACK confirms data, FIN ends it, RST resets it, PSH sends data immediately, URG marks urgent data.
Window Size	16	How much data the receiver can accept right now. It controls the speed of sending (flow control).
Checksum	16	Error check on the header and data. A damaged segment is discarded and sent again.
Urgent Pointer	16	Used with the URG flag to point to the urgent data inside the segment.
Options + Padding	Variable	Extra settings such as maximum segment size. Padding keeps the header a multiple of 32 bits.

Q4. Draw a diagram showing how a message sent from your device to Instagram's server passes through each TCP/IP layer, and annotate what happens at each step using simple app-related terms.

Ans:



Q5. Pick any one packet you captured in Wireshark and write which TCP/IP layer it belongs to, how you identified it, and what real-world action in an app it represents.

Ans:

Packet chosen	Details
Packet	TLSv1.3, Info: <i>Client Hello</i> (sent from my laptop to the Flipkart server on port 443)
TCP/IP layer	Application layer
How I identified it	In the Packet Details pane the headers were stacked as Frame, Ethernet II, IPv4, TCP and TLS. The TLS header was the top-most one, above TCP, so the packet carries application data. Expanding it also showed the server name <code>www.flipkart.com</code> in the Client Hello.
Real-world action	This is my browser starting a secure (HTTPS) session with Flipkart when I refreshed the homepage. It is the same first step that happens in an app before you view products, log in or pay for an order.